

Calculus I Lesson 30

- 1) A rectangle has a perimeter of 100 feet.

Find the length and width of a rectangle with a maximum area.

Let A = area Let x = length of rectangle Let y = width of rectangle

Note: $A = x \cdot y$; $P = 2x + 2y = 100$

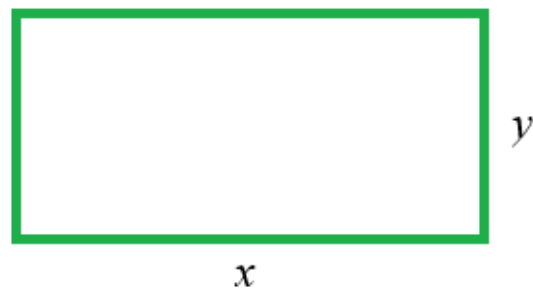
a) Solve for y : $2x + 2y = 100 \Rightarrow y = \underline{\hspace{1cm}} ?$

b) Write A as function of x only: $A = x \cdot y = \underline{\hspace{1cm}} ?$

c) Find $A' = \frac{dA}{dx} = \underline{\hspace{1cm}} ?$

d) Set $A' = 0$ and solve for x . $x = \underline{\hspace{1cm}} ?$

e) Area is a maximum when $x = \underline{\hspace{1cm}} ?$ and $y = \underline{\hspace{1cm}} ?$



2) A rectangle has an area of 80 square feet.

Find the length and width of a rectangle with a minimum perimeter.

Let P = perimeter Let x = length of rectangle Let y = width of rectangle

Note: $A = x \cdot y = 80$; $P = 2x + 2y$

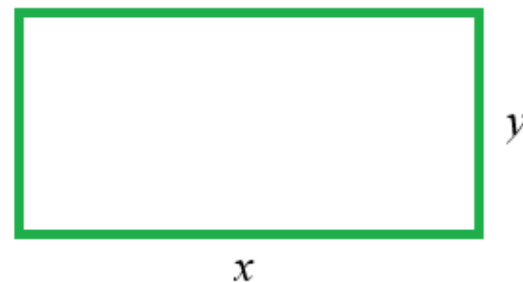
a) Solve for y : $x \cdot y = 80 \Rightarrow y = \underline{\hspace{1cm}} ?$

b) Write perimeter P as a function of x only: $P = 2x + 2y = \underline{\hspace{1cm}} ?$

c) Find $P' = \frac{dP}{dx} = \underline{\hspace{1cm}} ?$

d) Set $P' = 0$ and solve for x . $x = \underline{\hspace{1cm}} ?$

e) Perimeter is a minimum when $x = \underline{\hspace{1cm}} ?$ and $y = \underline{\hspace{1cm}} ?$



- 3) A container in the shape of a right circular cylinder with top and bottom has surface area 100 inch^2 .

What height (h) and base radius (r) will maximize the volume of the cylinder ?

$S = \text{Surface Area} = \text{Area of Top} + \text{Area of Bottom} + \text{Lateral Area}$

$$S = \pi r^2 + \pi r^2 + 2\pi rh$$

Note: $S = \pi r^2 + \pi r^2 + 2\pi rh = 100$

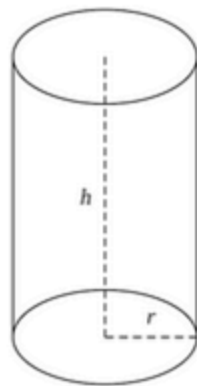
a) Solve $\pi r^2 + \pi r^2 + 2\pi rh = 100$ for h : $h = \underline{\hspace{1cm}} ?$

b) Write Volume $= \pi r^2 h$ as a function of r only: $V = \underline{\hspace{1cm}} ?$

c) Find $V' = \frac{dV}{dr} = \underline{\hspace{1cm}} ?$

d) Set $V' = \frac{dV}{dr} = 0$ and solve for r . $r = \underline{\hspace{1cm}} ?$

e) Cylinder is a maximum when $r = \underline{\hspace{1cm}} ?$ and $h = \underline{\hspace{1cm}} ?$



- 4) A sheet of cardboard 10 ft. by 15 ft. will be made into a box by cutting equal-sized squares from each corner and folding up the four edges. Each side of the cut-out square has side length of x . Find x so that the box has the largest volume ?

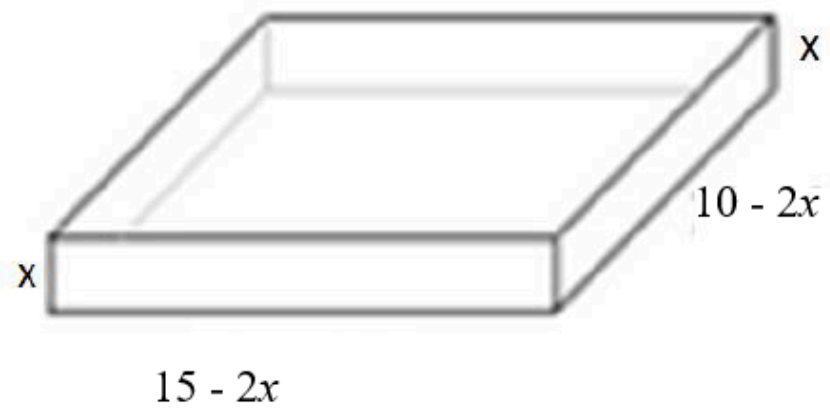
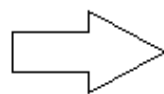
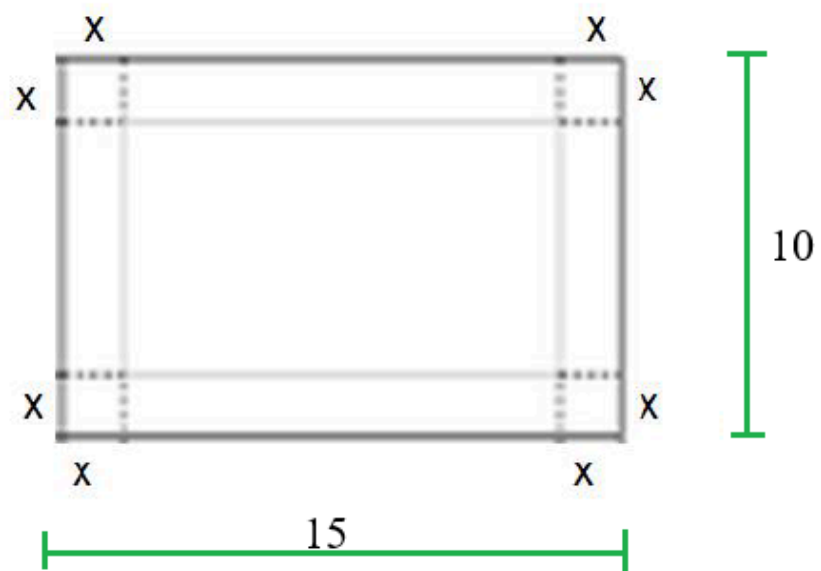
Let V be the volume of the box.

a) Write V as a function of x : $V = \underline{\hspace{1cm}}?$

b) $\frac{dV}{dx} = \underline{\hspace{1cm}}?$

c) Set $\frac{dV}{dx} = 0$ and solve for x . $x = \underline{\hspace{1cm}}?$

d) Largest volume of box = $\underline{\hspace{1cm}}?$



- 5) A manufacturer produces running shoes and the total daily cost of producing x pairs of shoes is: $C(x) = 100 + 20x - 0.5x^2$

How many pairs of shoes should the manufacturer produce in order to minimize production cost?

a) $\frac{dC}{dx} = \underline{\hspace{1cm}}?$

b) Set $\frac{dC}{dx} = 0$ and solve for x . $x = \underline{\hspace{1cm}}?$

Note: This is the number of pairs of shoes that will lead to minimized production cost.

6) A rancher is planning to build an enclosed area along a pond. It is to be rectangular with an area of 10,000 square feet and is to be fenced on the three sides not adjacent to the pond. Find the dimensions (x and y) such that the least amount of fencing that will be needed to construct the enclosed area?

a) $x + y + y = \underline{\hspace{1cm}}?$

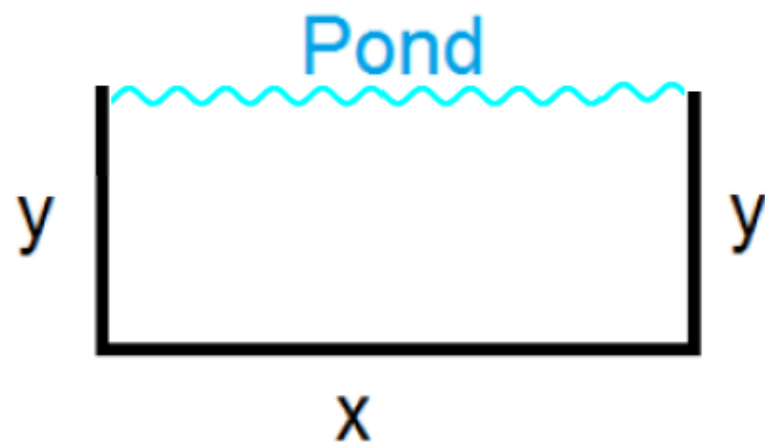
b) Solve the equation in part (a) for x in terms of y . $x = \underline{\hspace{1cm}}?$

c) $A =$ enclosed area $= x \cdot y$. Write A as a function of y only. $A = \underline{\hspace{1cm}}?$

d) $\frac{dA}{dy} = \underline{\hspace{1cm}}?$

e) Set $\frac{dA}{dy} = 0$ and solve for y . $y = \underline{\hspace{1cm}}?$

f) Use the equation in part (a) to find x . $x = \underline{\hspace{1cm}}?$



7) A baseball is hit 3 feet from the ground level with speed of 120 feet/sec and angle of elevation is $\theta = 30^\circ = \pi / 6$ rad.

Find the maximum height reached by the ball.

Equation for the ball's path:

$$y = h + v_0 \sin \theta \left[\frac{x}{v_0 \cos \theta} \right] - \frac{1}{2} \cdot g \left[\frac{x}{v_0 \cos \theta} \right]^2; \quad g = \text{Earth's gravity} = 32 \text{ ft/sec}^2$$

$$y = 3 + 120 \sin(\pi/6) \left[\frac{x}{120 \cos(\pi/6)} \right] - \frac{1}{2} \cdot 32 \left[\frac{x}{120 \cos(\pi/6)} \right]^2$$

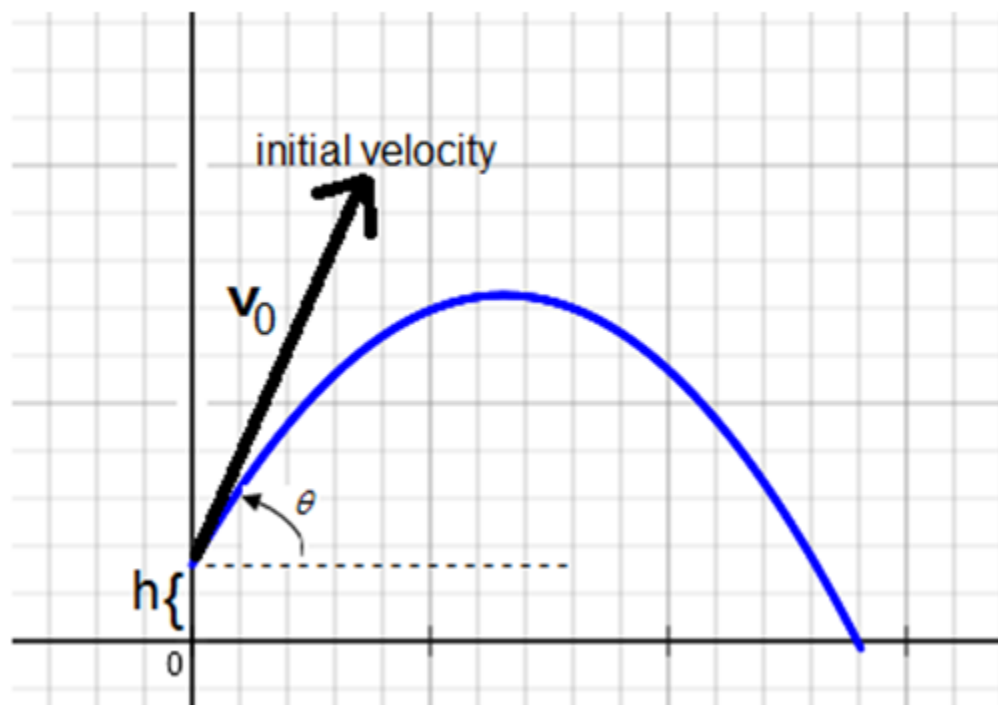
$$y = 3 + 60 \left[\frac{x}{60\sqrt{3}} \right] - \frac{1}{2} \cdot 32 \left[\frac{x}{60\sqrt{3}} \right]^2 = 3 + \frac{1}{\sqrt{3}} x - \frac{16}{(60\sqrt{3})^2} x^2$$

$$y = 3 + 0.57735x - 0.00148148148x^2$$

a) $y' =$ _____ ?

b) Set $y' = 0$ and solve for x . $x =$ _____ ?

c) Ball reached maximum height at the point: $x =$ _____ ? and $y =$ _____ ?



- 8) A cannon ball is fired 10 feet from the ground level with speed of 100 feet/sec and angle of elevation is $\theta = 45^\circ = \pi/4$ rad.

Find the maximum height reached by the cannon ball.

Equation representing the cannon ball's path is:

$$y = h + v_0 \sin \theta \left[\frac{x}{v_0 \cos \theta} \right] - \frac{1}{2} \cdot g \left[\frac{x}{v_0 \cos \theta} \right]^2; \quad g = \text{Earth's gravity} = 32 \text{ ft/sec}^2$$

$$y = 10 + 100 \sin(\pi/4) \left[\frac{x}{100 \cos(\pi/4)} \right] - \frac{1}{2} \cdot 32 \left[\frac{x}{100 \cos(\pi/4)} \right]^2$$

$$y = 10 + 50\sqrt{2} \left[\frac{x}{50\sqrt{2}} \right] - 16 \left[\frac{x}{50\sqrt{2}} \right]^2 = 10 + x - \frac{16}{5000} x^2$$

a) $y' =$ _____ ?

b) Set $y' = 0$ and solve for x . $x =$ _____ ?

c) Ball reached maximum height at the point: $x =$ _____ and $y =$ _____

